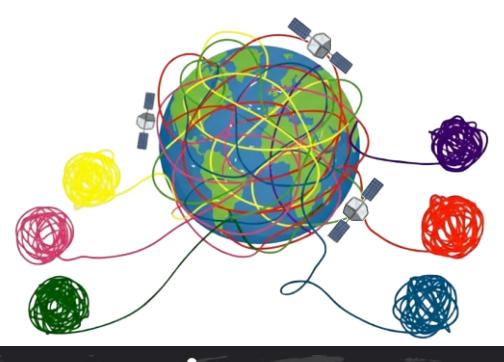




Mixture of Geographical Experts: Disentangling Earth



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Background and Motivation

Background

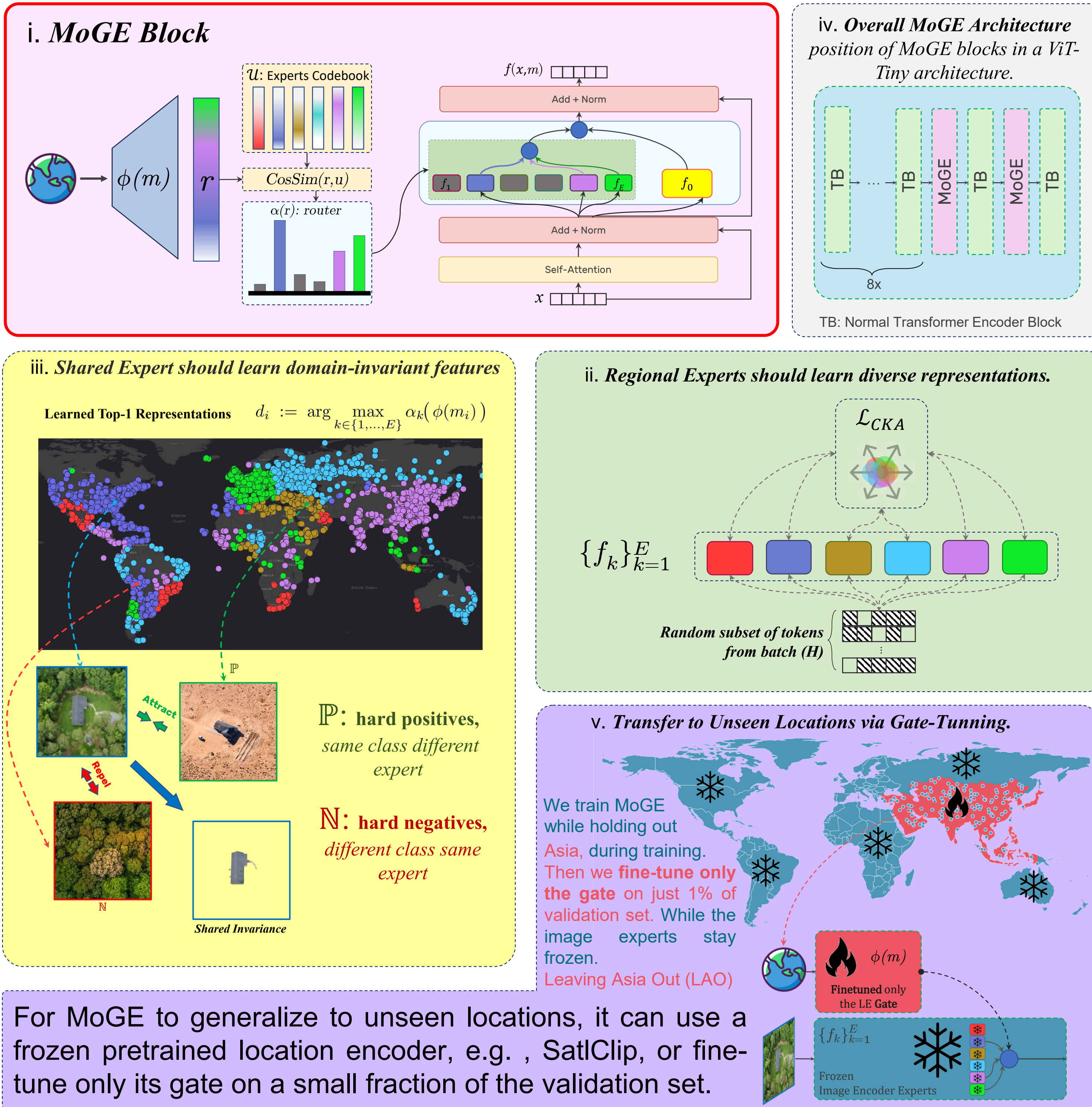
- Geographical confounders **break the covariate-shift assumption** in EO tasks, causing Domain-Invariant methods to fail to generalize globally.
- Domain Generalization (DG) methods rely on **manually defined domains**, with **hard boundaries**, that need **expert knowledge** for each task.

Motivation, Following the First Law of Geography*:

- Location** can be used to **learn** soft overlapping **domains** from the data itself.
- It is also a **proxy** to condition on unobserved confounders.

*According to Tobler: "Everything is related to everything else, but near things are more related than distant things."

MoGE: Mixture of Geographical Experts



Contributions

In this work we introduce **MoGE**, a geo-routed Mixture-of-Experts that:

- Learns concept-consistent geographic domains** from location metadata, removing the need for hand-crafted domain definitions.
- Disentangles global invariance from spatial variation**, yielding strong improvements over both DG and domain-specific (DS) baselines.
- Plugs into any Transformer layer **without altering pretrained weights**.
- A novel method to mitigate **spurious Location-Class correlations**.

- i. MoGE replaces the FFN in a Transformer encoder block with a geo-routed MoE composed of one shared invariant expert and several specialized experts selected through metadata-driven routing.

$$f(x, m) = \underbrace{\gamma f_0(x)}_{\text{invariant}} + \underbrace{(1 - \gamma) \sum_{k=1}^E \alpha_k(\phi(m)) f_k(x)}_{\text{specialized}}$$

- ii. To enforce **each expert to learn distinct and orthogonal representations**, we apply a CKA-based diversity loss that penalizes similarity between their outputs.

$$\mathcal{L}_{CKA} = \frac{2}{E(E-1)} \sum_{1 \leq i < j \leq E} \text{CKA}_{\text{lin}}(H_i, H_j) \quad H_k = f_k(H) \text{ for } k = 1, \dots, E$$

- iii. To have an invariant fallback, we force the shared expert features (f_0) to be **invariant across domains that were discovered** by our location encoding gating function.

$$\mathcal{L}_{SupCon} = \frac{-1}{B} \sum_{i \in \mathcal{A}} \left[\log \sum_{p \in \mathbb{P}_i} e^{\langle q_i, p \rangle / \tau_{sup}} - \log \sum_{v \in \mathbb{P}_i \cup \mathbb{N}_i} e^{\langle q_i, v \rangle / \tau_{sup}} \right] \quad \begin{aligned} q_i &:= f_0(x_i) / \|f_0(x_i)\|_2 \\ \mathcal{A} &:= \{i \text{ in batch} : |\mathbb{P}_i| \geq 1\} \\ B &:= |\mathcal{A}| \end{aligned}$$

- iv. Probabilistic Group Robustness: To prevent biased experts.

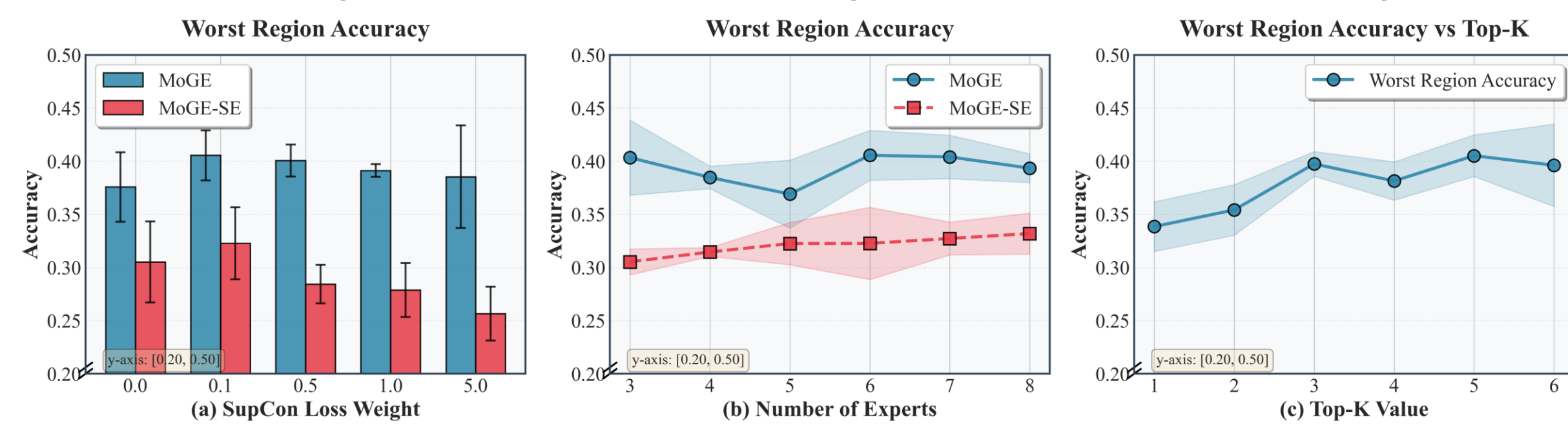
$$\mathcal{L}_{PGDRO} = \max_{(y,k) \in \mathcal{G}} \left[\frac{1}{\tilde{n}_{y,k}} \sum_{i: y_i=y} \pi_{i,k} \ell_i + \frac{C}{\sqrt{\tilde{n}_{y,k}}} \right] \quad \begin{aligned} \pi_{i,k} &: \text{gate probability of sample } i \text{ for expert } k \\ \tilde{n}_{y,k} &:= \sum_{i: y_i=y} \pi_{i,k} \\ \mathcal{G} &:= \mathcal{Y} \times \{1, \dots, E\} \\ C &\geq 0 : \text{group-size adjustment} \end{aligned}$$

- **The total MoGE loss** is given below, where $\mathcal{L}_{aux}$ serves as a load-balancing term that encourages an even distribution of routing assignments.

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{CKA} \mathcal{L}_{CKA} + \lambda_{SupCon} \mathcal{L}_{SupCon} + \lambda_{DRO} \mathcal{L}_{PGDRO}$$

Ablation Studies

- Ablation 1: Effect of Invariance loss, number of experts, and top-k routing.** Using 6 experts and top-3 routing yields the best results. Furthermore, using invariance loss improves both shared expert and overall MoGE performance.

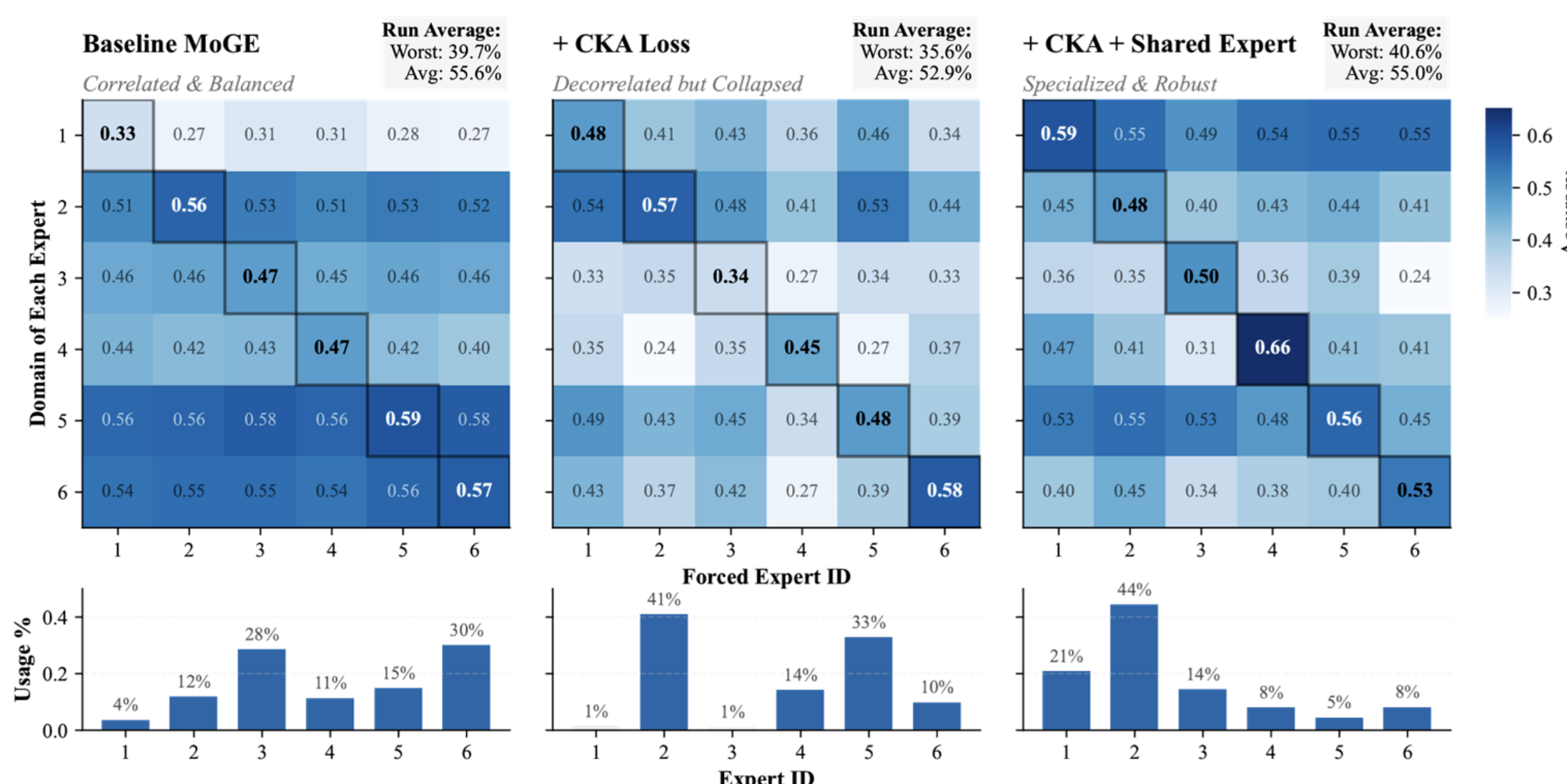


- Ablation 2: Effect of Diversity loss and shared expert.** Without the CKA loss, experts remain highly correlated. Adding CKA without the shared expert decorrelates them but leads to expert collapse and reduced performance. Using both together avoids collapse and gives the best results.

In the matrix:

Each **row** corresponds to an **expert's domain**, defined as the samples for which that expert is the top-1 route.

Each **column** shows the performance of an expert when all the samples from a given domain are forced through it.

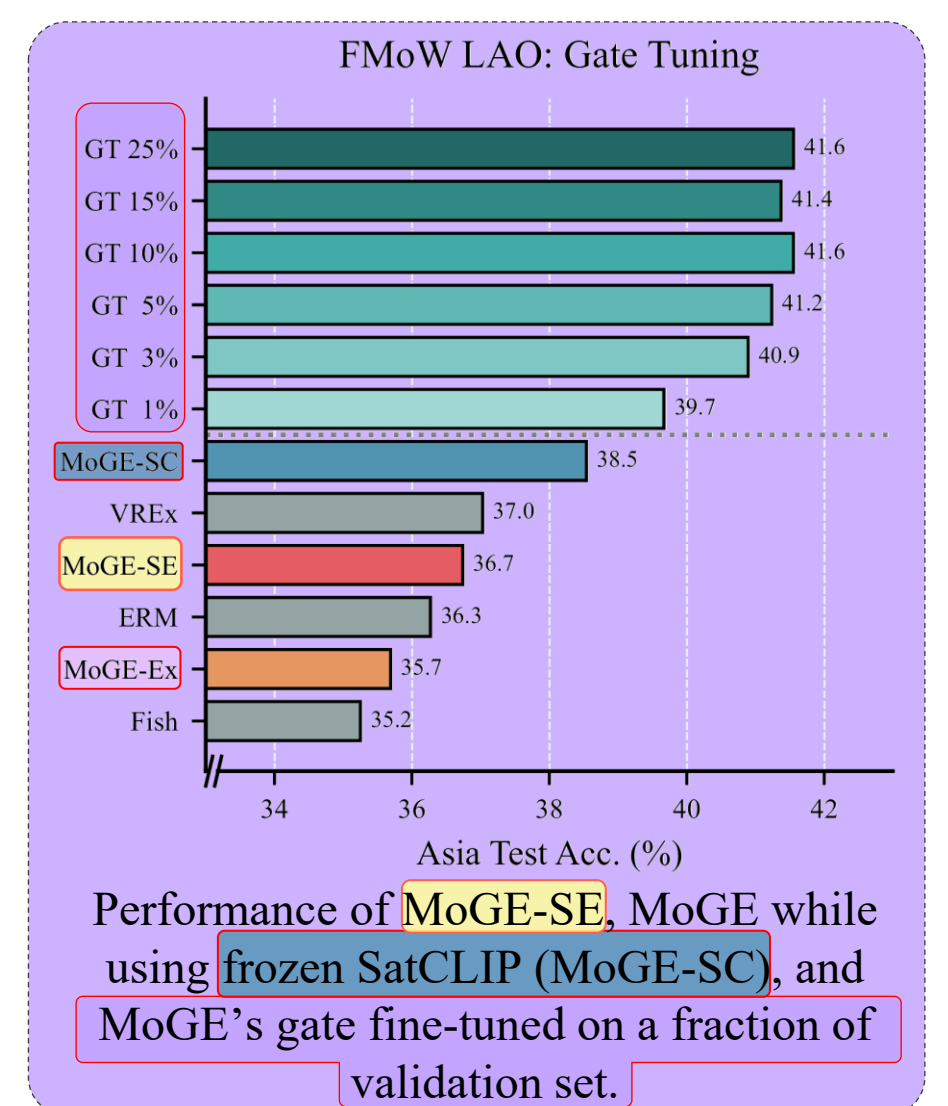


Experimental Results

- MoGE** achieves the **best** results on **FMoW** and **MapInWild** datasets.
- The **shared expert alone (MoGE-SE)** also outperforms DG methods.
- Gate-Tuning** on unseen locations, using only 150 samples, from Asia, outperforms all DG method.

Method	FMoW (62 classes and 5 regions as domains)			MapInWild (2 classes, 5 regions resulting in 10 region-class groups)		
	WRA	Overall Acc.		WGA	WCA	Overall Acc.
Domain Specific						
MoGE	38.85 ± 2.50	55.25 ± 0.57		60.91 ± 1.38	74.20 ± 1.51	77.20 ± 0.53
ERM+LE	35.83 ± 0.51	53.23 ± 0.72		48.76 ± 9.12	70.58 ± 3.43	74.48 ± 1.46
D3G	33.38 ± 0.64	51.51 ± 0.19		52.02 ± 5.38	68.16 ± 4.01	77.73 ± 0.79
D3G+LE	34.60 ± 1.27	50.76 ± 0.12		55.16 ± 6.33	70.01 ± 5.31	77.82 ± 1.25
DCP+LE	35.97 ± 3.98	52.30 ± 1.08		51.39 ± 7.98	64.43 ± 4.91	75.42 ± 1.46
Per Region Models	26.75 ± 1.26	49.72 ± 0.26		50.11 ± 5.41	70.68 ± 1.80	76.49 ± 0.56
Domain Invariant						
MoGE-SE	33.01 ± 0.34	48.60 ± 0.41		57.83 ± 4.37	74.32 ± 1.07	77.50 ± 0.96
ERM	32.43 ± 1.67	53.69 ± 0.37		52.18 ± 4.38	70.40 ± 3.43	77.16 ± 1.73
GroupDRO	30.70 ± 0.80	49.06 ± 0.37		57.25 ± 0.75	73.87 ± 1.05	75.70 ± 0.75
IRM	25.85 ± 0.93	42.71 ± 0.41		54.81 ± 2.98	72.94 ± 1.10	75.57 ± 1.23
Fish	33.08 ± 0.29	44.99 ± 0.72		52.02 ± 1.85	69.20 ± 1.14	76.72 ± 1.39
Mixup	32.88 ± 0.63	47.25 ± 0.62		55.52 ± 0.95	71.75 ± 1.60	75.40 ± 0.92
VREx	32.48 ± 1.28	46.83 ± 0.90		56.76 ± 0.63	70.99 ± 0.48	75.57 ± 0.48
RDM	32.66 ± 0.60	47.70 ± 0.17		55.60 ± 0.43	70.15 ± 1.25	73.44 ± 0.96

WRA: Worst Region Acc. | WGA Worst Group Acc. | WCA: Worst class Acc.



Discussion

- MoGE brings together the strengths of both DG and DS methods and **delivers significant improvements on both fronts**, demonstrating that *learning domains directly from data* can outperform handcrafted domain definitions.
- Beyond performance, **MoGE also offers clearer interpretability** compared to a naïve use of location as an input, and avoids location-class biases.
- MoGE offers Parameter Efficient Fine-Tuning (PEFT) by **Gate-Tuning** for unseen regions.



Code on GitHub



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